✈️ Air Tracker Dashboard

Project Overview

Air Tracker Dashboard is a flight analytics web application developed using Python, SQLite, and Streamlit. The project collects aviation data, stores it in a relational database, performs SQL-based analysis, and presents insights through an interactive dashboard.

This project was developed as part of a Data Analytics capstone project focusing on API Integration, SQL Database Management, Data Analysis, and Streamlit Dashboard Development.

Features

📊 Dashboard Metrics

- Total Airports
- Total Aircraft
- Total Flights
- Total Delay Records

🔍 Flight Search

- Search flights by Flight Number
- View flight details instantly

🛫 Airport Information

- View airport details
- Airport codes, city, and country information

📈 Flight Status Analysis

- Analyze flight statuses
- Visual representation using charts

🏆 Top Routes Analysis

- Identify busiest flight routes
- Route-wise flight counts

⏱ Airport Delay Analysis

- Analyze airport delay statistics
- Average delay trends
- Arrival and departure delay comparisons

Technologies Used

- Python
- SQLite
- Streamlit
- Pandas
- Requests
- AeroDataBox API (RapidAPI)

Database Schema

Airport Table

Stores airport information:

- Airport ID
- IATA Code
- ICAO Code
- Airport Name
- City
- Country

Aircraft Table

Stores aircraft details:

- Registration
- Model
- Manufacturer
- ICAO Type Code
- Owner

Flights Table

Stores flight information:

- Flight Number
- Airline Code
- Origin Airport
- Destination Airport
- Flight Status
- Aircraft Code

Airport Delays Table

Stores delay analytics:

- Airport IATA
- Delay Date
- Average Delay
- Arrival Delay
- Departure Delay

Project Structure

AirTracker-Dashboard/

├── app.py

├── airtracker.db

├── AirTracker_Final_Project.ipynb

├── requirements.txt

├── README.md

└── screenshots/

Installation

Clone Repository

```
git clone https://github.com/your-username/AirTracker-Dashboard.git
```

Navigate to Project Folder

```
cd AirTracker-Dashboard
```

Install Dependencies

```
pip install -r requirements.txt
```

Run Application

```
streamlit run app.py
```

Sample Dashboard Modules

- Dashboard Overview
- Flight Search
- Airport Information
- Flight Status Analysis
- Top Routes
- Airport Delay Analysis

Learning Outcomes

- API Integration using AeroDataBox API
- JSON Data Processing
- SQLite Database Design
- SQL Query Development
- Data Analytics
- Interactive Dashboard Development using Streamlit

Future Enhancements

- Real-Time Flight Tracking
- Airport Weather Integration
- Flight Delay Prediction using Machine Learning
- Interactive Maps and Geospatial Analysis

Author

Hari Krishna

Data Analytics & Python Development Project